

# Demonstrating Diesmann 1999

exp087 · 20 August 2026 · Draft

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## Abstract

A synfire chain is a sequence of neuron pools connected in one direction. A brief packet of nearly synchronous spikes enters the first pool, which may generate a new packet in the next pool. Diesmann et al. (1999) showed that weak or diffuse packets disappear, while sufficiently strong packets can converge toward a stable size and temporal width [1]. At the selected SNNLANG operating point, a weak packet disappears after two pools. A broad packet grows and sharpens, while an oversized narrow packet relaxes immediately. Both surviving packets reach the same 100-spike, approximately 0.06 ms state. A grid of initial packet sizes and widths separates extinction from propagation. This qualitatively demonstrates the stable-packet attractor described by Diesmann et al.

## Prior work

Diesmann et al. identified two regimes separated by a boundary in packet state space. Below the boundary, activity dies. Above it, different input packets approach the same stable packet [1]. Our neuron model, background drive, and finite chain differ from the original model, so this is a qualitative demonstration rather than an exact reproduction.

## Methods

**Methods 1–4 are complete. Every simulation uses one network seed and one fixed background realization.**

1. **Define the chain and the pulse packet.** Build 6 feedforward pools of 100 conductance-based leaky integrate-and-fire neurons. Each neuron receives exactly 40 excitatory connections from the preceding pool, with total incoming strength 0.33  $\mu\text{S}$  and delay 1 ms. Independent background spikes keep neurons near firing threshold without producing population volleys by themselves. The first pool receives an artificial pulse packet through the same feedforward connection rule.

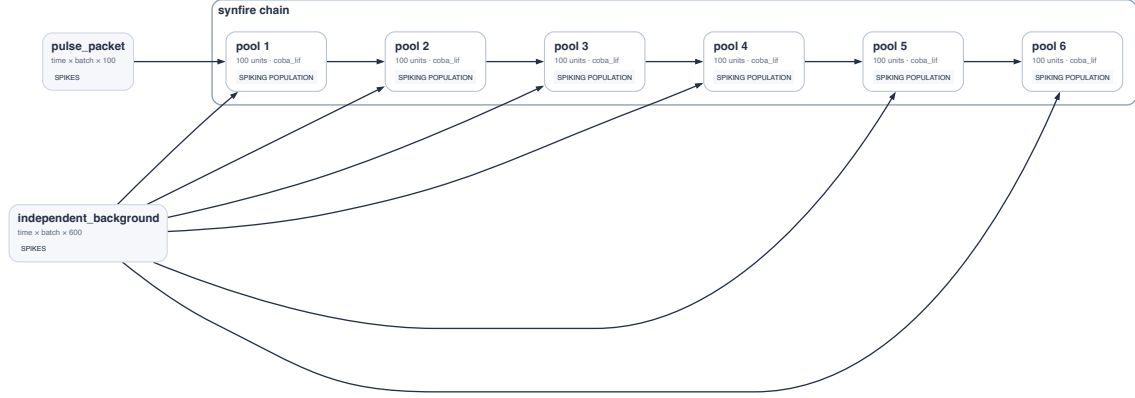


Figure 1: SNNLANG circuit export. A pulse packet enters Pool 1 and activity can travel through six feedforward pools. Independent background input reaches every pool.

Describe each packet by its size  $\alpha$ , the number of input spikes, and its temporal spread  $\sigma$ , the standard deviation of their spike times. A narrow packet concentrates excitation and is more likely to make the target pool fire. A broad packet distributes the same input over time and may fail to cross threshold.

#### A pulse packet has a size and a temporal width

Design schematic:  $\alpha$  counts spikes;  $\sigma$  measures their spread around the packet centre

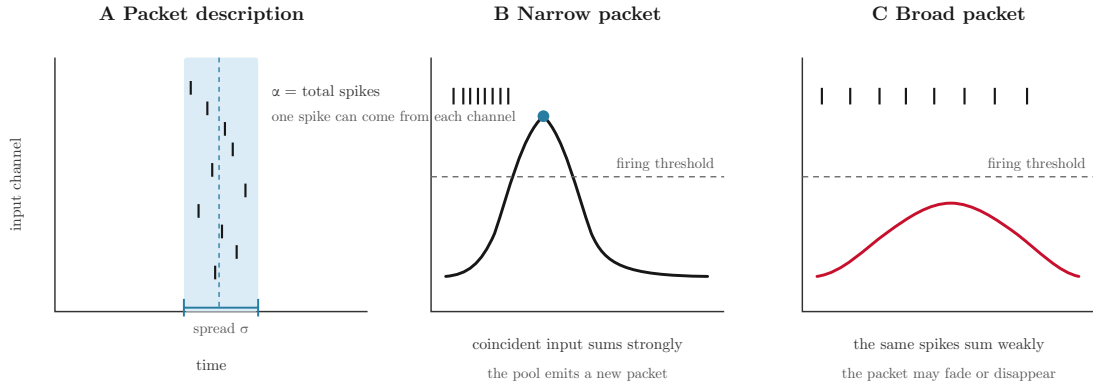


Figure 2: **Pulse packet definition.** Packet size  $\alpha$  counts spikes. Packet spread  $\sigma$  measures their timing dispersion. The curves illustrate why equal-sized packets need not have the same effect. They are not simulated data.

Generates Result 1.

2. **Compare packets with different starting states.** Use one fixed realization of the background input. Test feedforward strengths 0.33, 0.35, 0.37, and 0.40  $\mu\text{S}$  at background rates 2, 5, and 10 Hz. Select the weakest point at which a 50-spike, 5 ms reference packet reaches every pool without secondary volleys. Freeze those network settings. Then test a weak diffuse packet, a broad strong packet, and a narrow oversized packet. For every pool, count the spikes in its response volley to obtain  $\alpha$ , and calculate their spike-time standard deviation to obtain  $\sigma$ .

### Packets either die or approach the stable packet

Design schematic: two different propagating packets should converge toward the same state

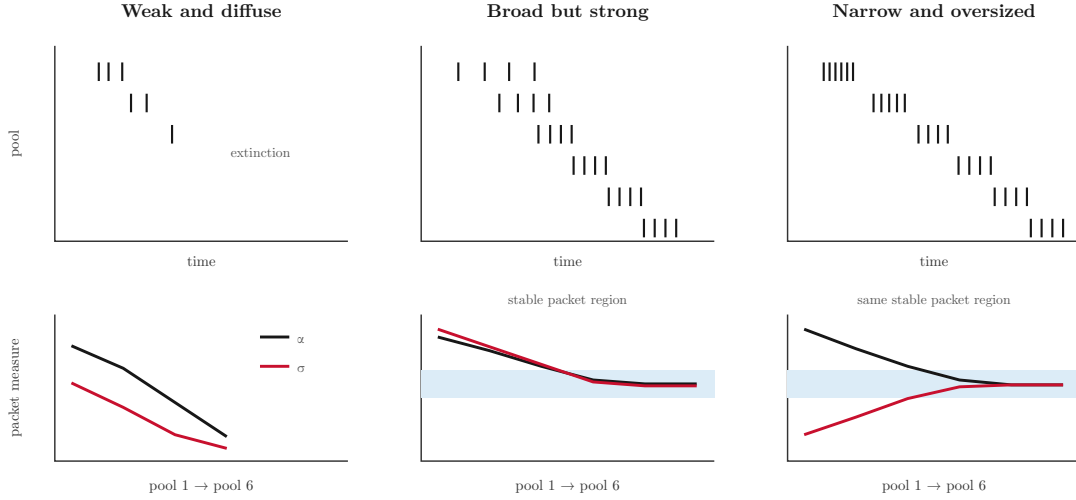


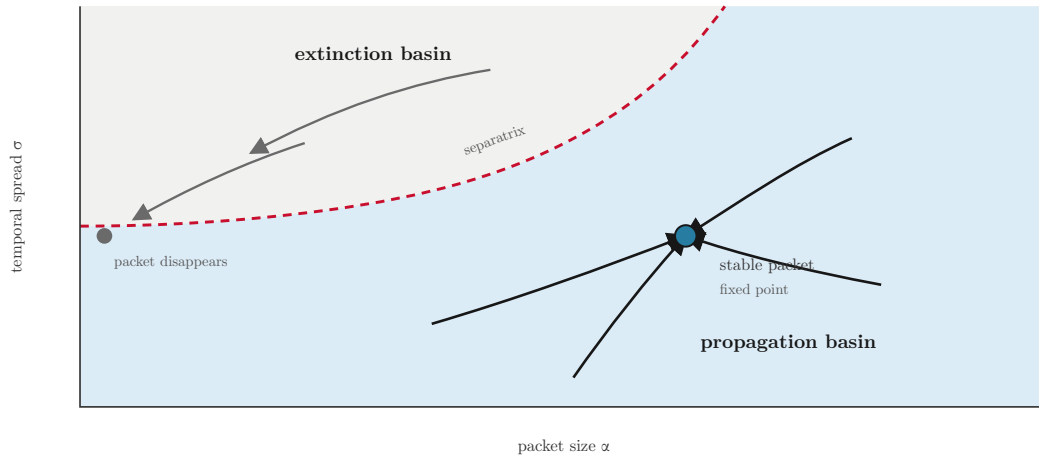
Figure 3: **Expected packet fates.** A weak or diffuse packet should die. Two different packets inside the propagation regime should approach the same stable size and spread. These trajectories explain the comparison and are not simulated data. Result 2 reproduces this layout with measured spikes and packet statistics.

Generates Result 2.

3. **Map the packet state transformation.** Keep the tuned network and background input fixed. Test 11 packet sizes from 20 to 90 spikes and seven temporal spreads from 0.2 to 5 ms. At each pool, plot the measured packet state  $(\alpha_n, \sigma_n)$  and draw an arrow to the state produced in the next pool,  $(\alpha_{n+1}, \sigma_{n+1})$ . Classify a packet as extinct when no later pool produces a response volley. Locate the boundary between extinction and propagation, and test whether surviving trajectories approach one common state.

### The chain should contain an extinction boundary and a packet attractor

Design schematic: arrows show the layer-to-layer transformation of packet state



Result 3 will replace this schematic with measured layer-to-layer vectors.

Figure 4: **Expected packet state space.** The dashed line separates initial packets that disappear from those that propagate. Layer-to-layer arrows in the propagation basin should approach one stable packet. This is a design schematic, not simulated data. Result 3 shows the measured transformations.

Generates Result 3.

4. **Render a looping raster hero.** Use the measured spike times from the three packet trajectories in Method 2. Place neurons from Pools 1–6 in separate horizontal bands, with time moving horizontally. Scroll the raster at a constant speed. Show the weak packet fading out and the two surviving packets approaching the same compact volley. Incoming spikes must stop at each pool and a newly generated volley must appear in the next pool. Repeat the complete sequence at an exact interval and match the first and last frames to produce an eight-second seamless MP4 loop.

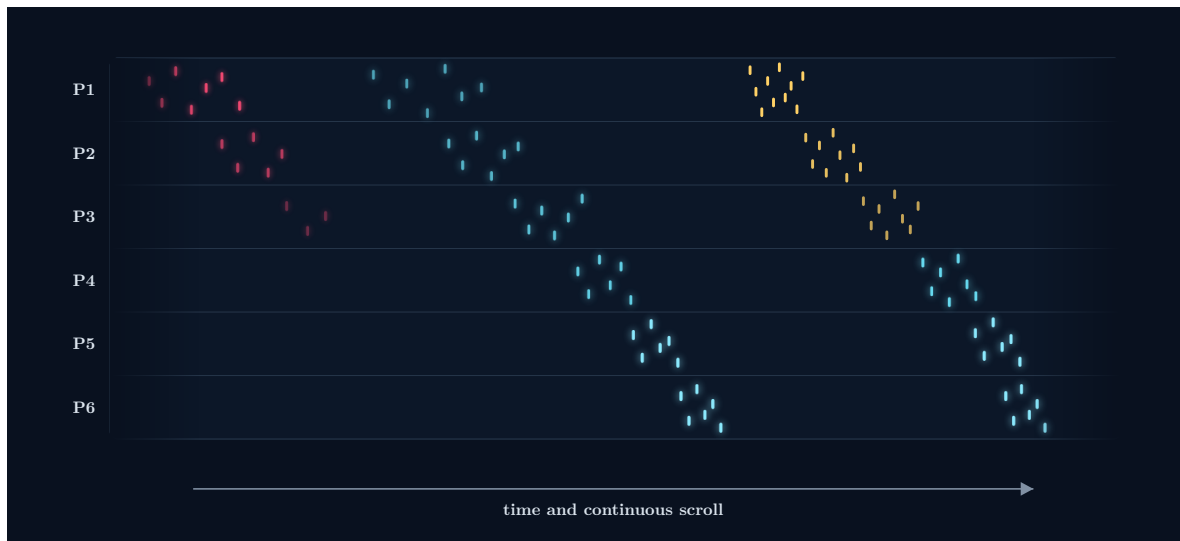


Figure 5: **Approximate frame from the hero animation.** Result 4 replaces these illustrative marks with measured spikes. Packet size appears as spike density and packet spread appears as horizontal volley width.

Generates Result 4.

## Results

### 1. Reference packet propagation.

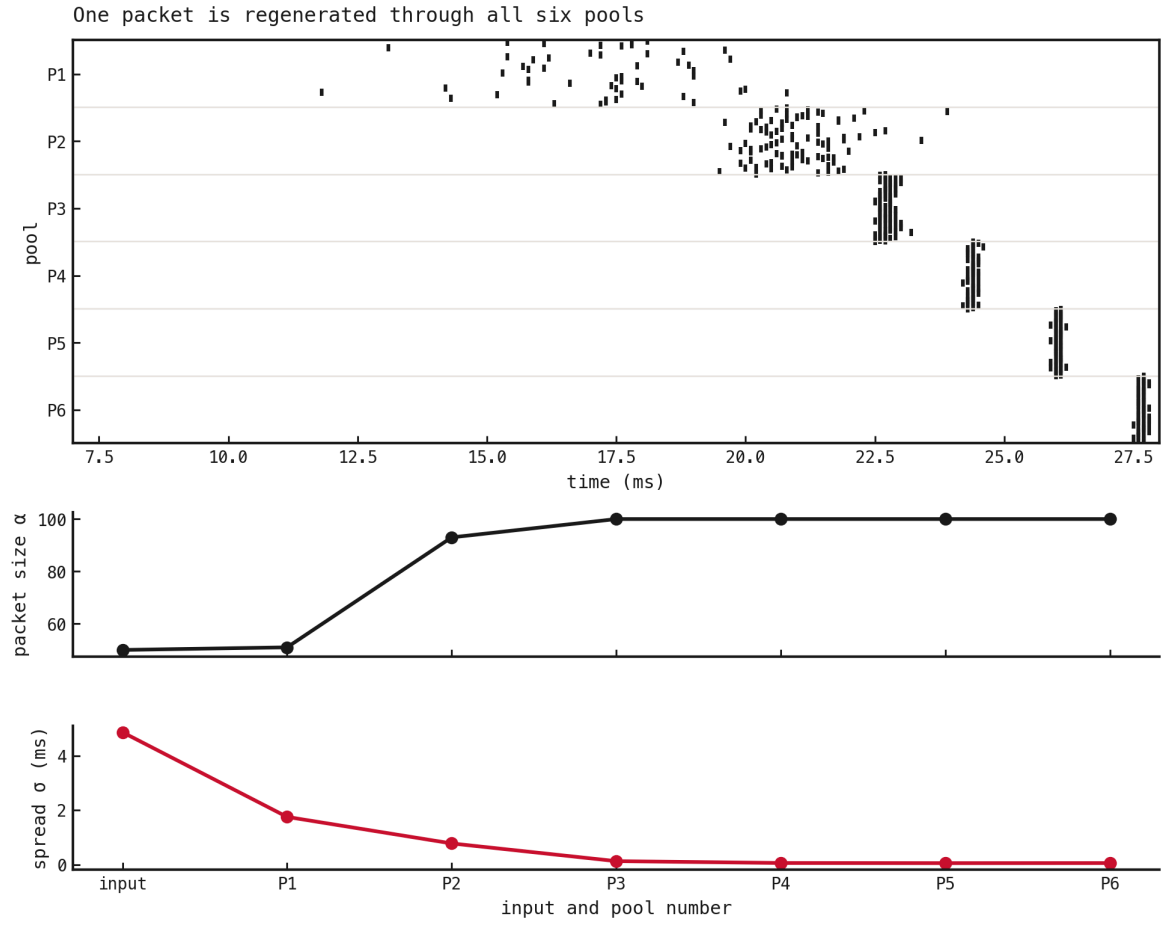


Figure 6: Reference-packet propagation at total feedforward strength  $0.33 \mu\text{S}$  and background rate 10 Hz.

The broad reference packet produces 51 spikes in Pool 1 and 93 in Pool 2. It then reaches all 100 neurons while narrowing from 1.75 ms to approximately 0.06 ms. Each pool emits one regenerated volley.

### 2. Three packet trajectories.

Measured packets either die or approach the stable packet

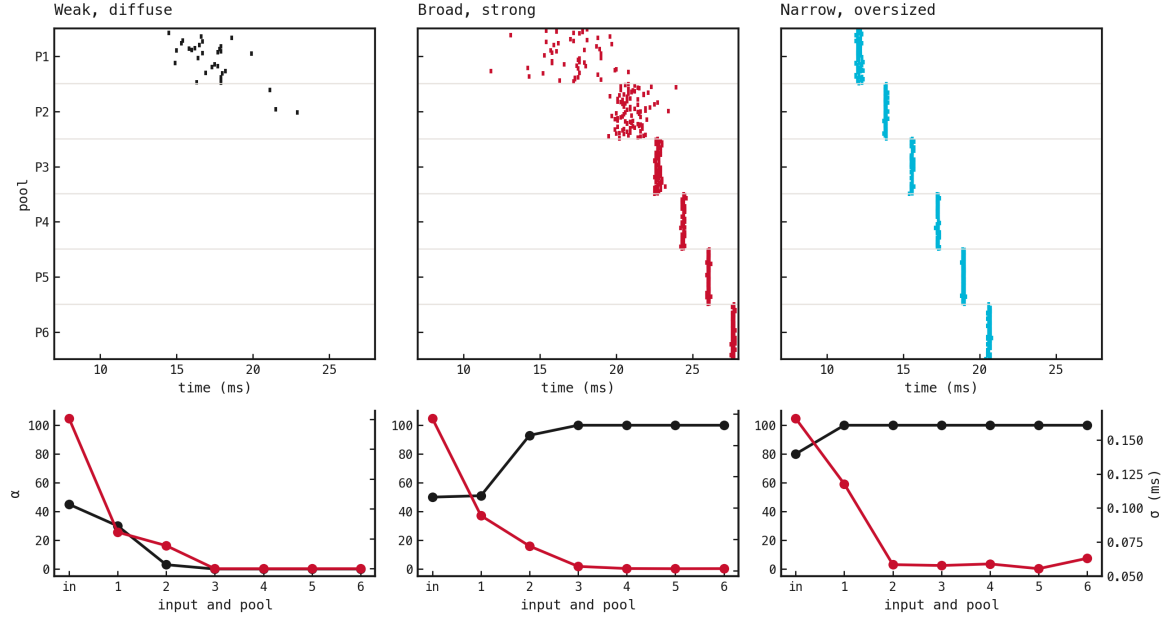


Figure 7: Measured packet fates. The upper panels show spikes grouped by pool. The lower panels show packet size  $\alpha$  in black and spread  $\sigma$  in red.

The weak packet falls from 30 spikes to 3, then disappears. The broad and oversized packets both reach 100 spikes with late-chain spreads of 0.06 and 0.06 ms. Different initial packets therefore converge on the same measured state.

### 3. Packet state space.

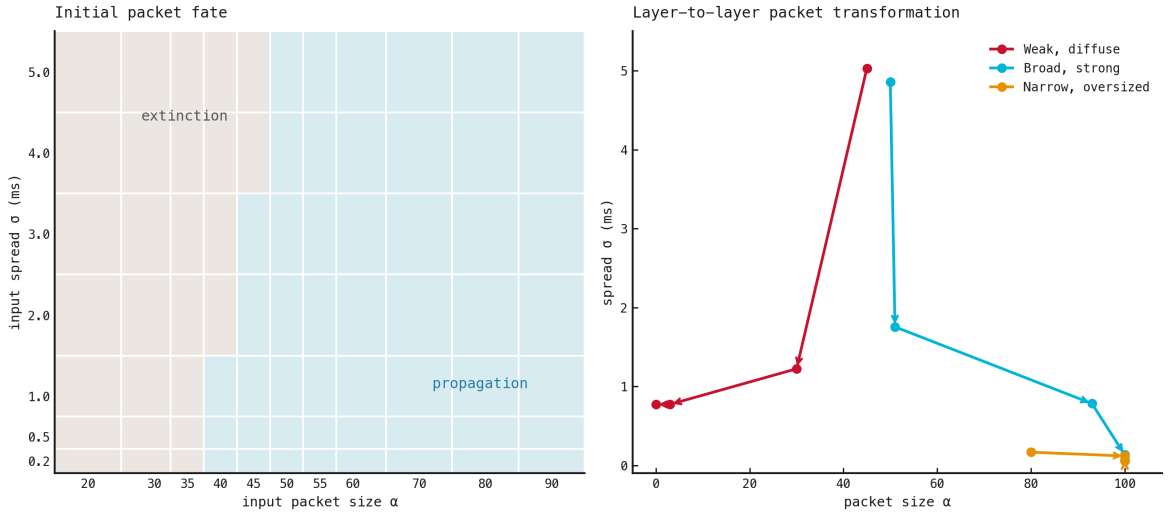


Figure 8: Measured packet state space. Tiles classify 77 initial size-width combinations. Arrows follow the three representative packets from their inputs through successive pools. 50 of 77 tested packets reach Pool 6. The boundary shifts toward larger  $\alpha$  as  $\sigma$  increases: temporally diffuse packets require more spikes. Surviving trajectories collapse toward the saturated 100-spike state, while the weak trajectory moves toward extinction.

### 4. Looping raster hero.

[ Video — Measured spikes from the three packet trajectories, arranged as an eight-second seamless scrolling raster. · view the web edition to play. ]

Red shows the packet that dies. Cyan shows the broad packet sharpening as each pool regenerates it. Amber begins as the oversized packet and becomes cyan after it reaches the same stable form. Spike marks fade at the horizontal boundaries so the periodic time window loops without a visible cut.

## References

### References

1. Diesmann, M., Gewaltig, M.-O., and Aertsen, A. (1999). Stable propagation of synchronous spiking in cortical neural networks. *Nature* 402, 529–533. doi:10.1038/990101